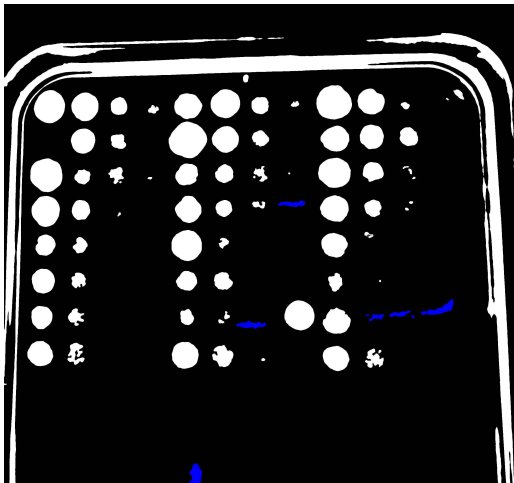


Msm-1_NoDrug_IMG_3319.JPG Log

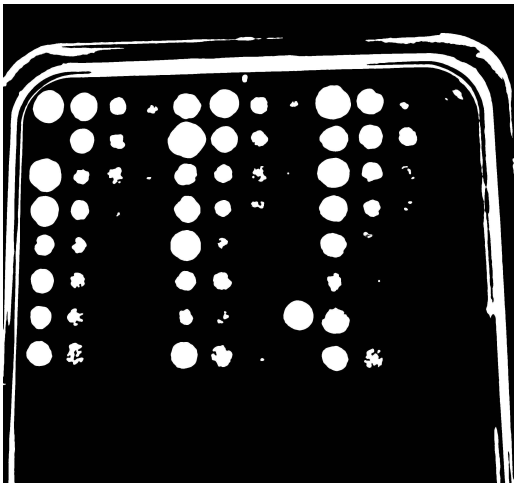


Filename: Msm-1_NoDrug_IMG_3319.JPG
Additive: None
X Dilution: 10
Y Dilution: 2
Strains: Msm, Msm, Msm
Column Indexes: [[0, 1, 2, 3], [4, 5, 6, 7], [8, 9, 10, 11]]
Gap Between Strains: False
Threshold: 16.0
Minimum Area: 15
Block Size: 197

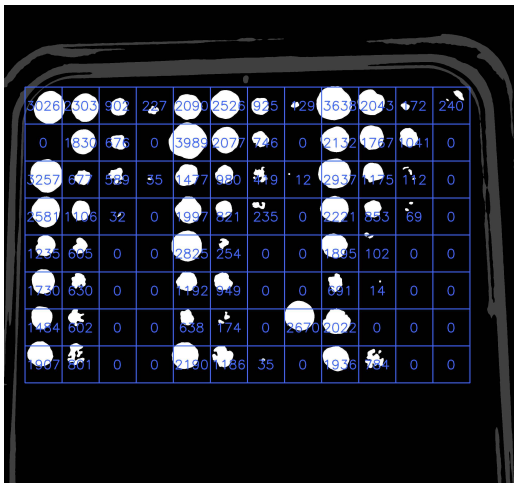
Preview Image



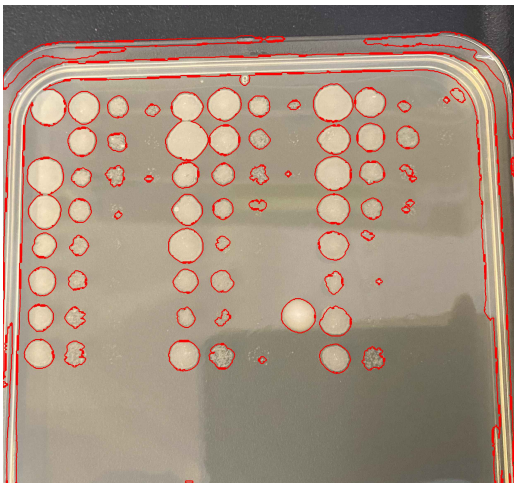
Tool Usage: Red(+) Blue(-)



Binary Image

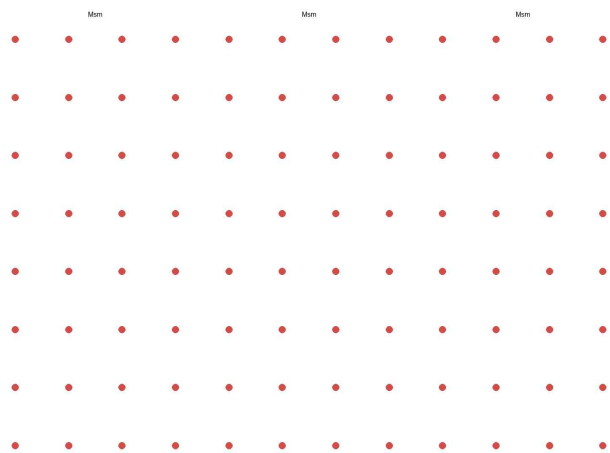


Grid Image



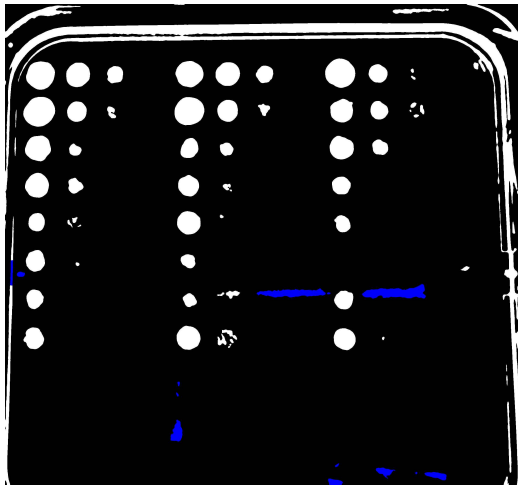
Contours Image

Msm-2_Kanamycin_IMG_3323.JPG Log

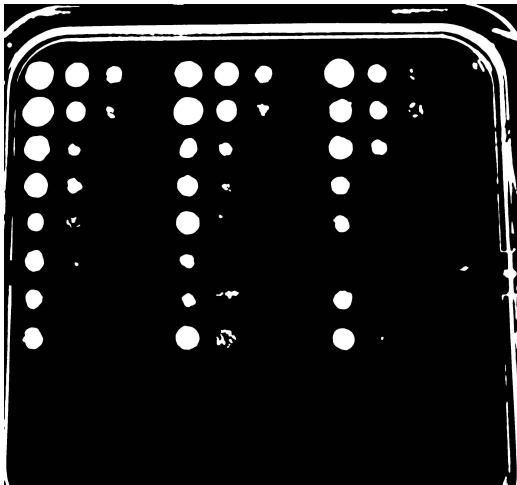


Filename: Msm-2_Kanamycin_IMG_3323.JPG
Additive: Kanamycin
X Dilution: 10
Y Dilution: 2
Strains: Msm, Msm, Msm
Column Indexes: [[0, 1, 2, 3], [4, 5, 6, 7], [8, 9, 10, 11]]
Gap Between Strains: False
Threshold: 17.0
Minimum Area: 15
Block Size: 197

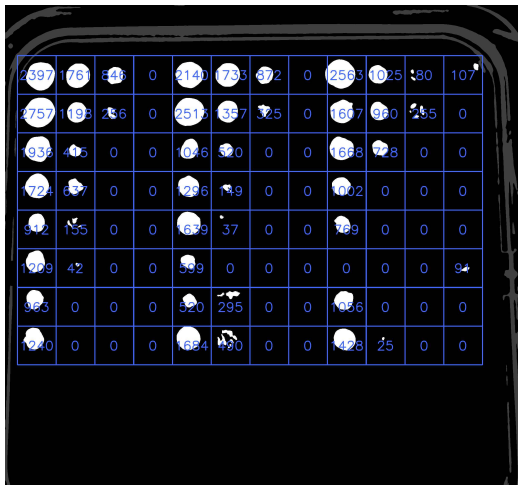
Preview Image



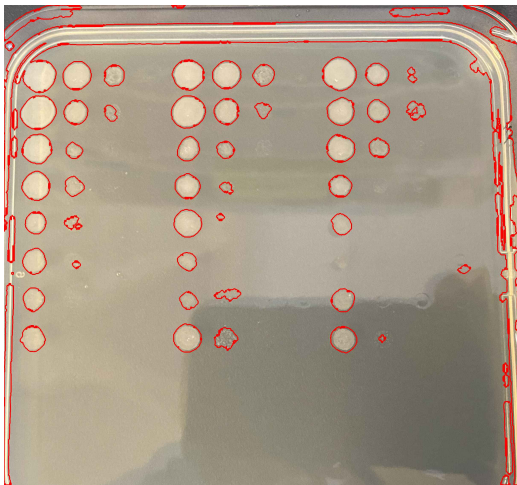
Tool Usage: Red(+) Blue(-)



Binary Image

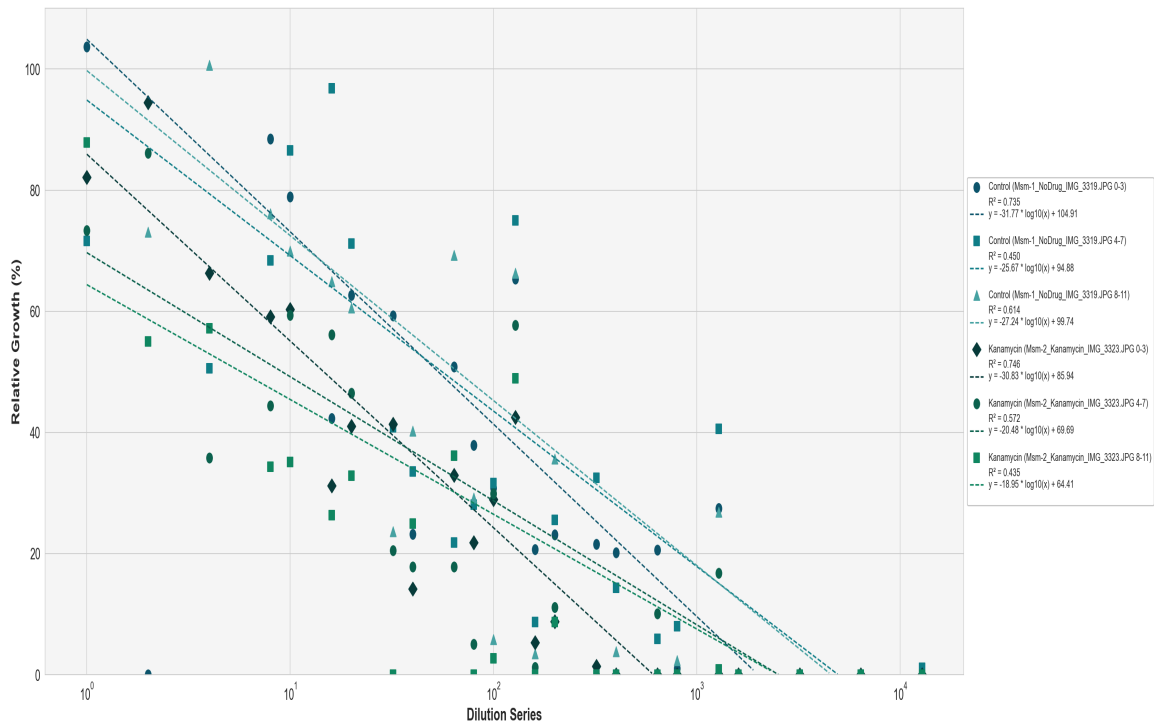


Grid Image



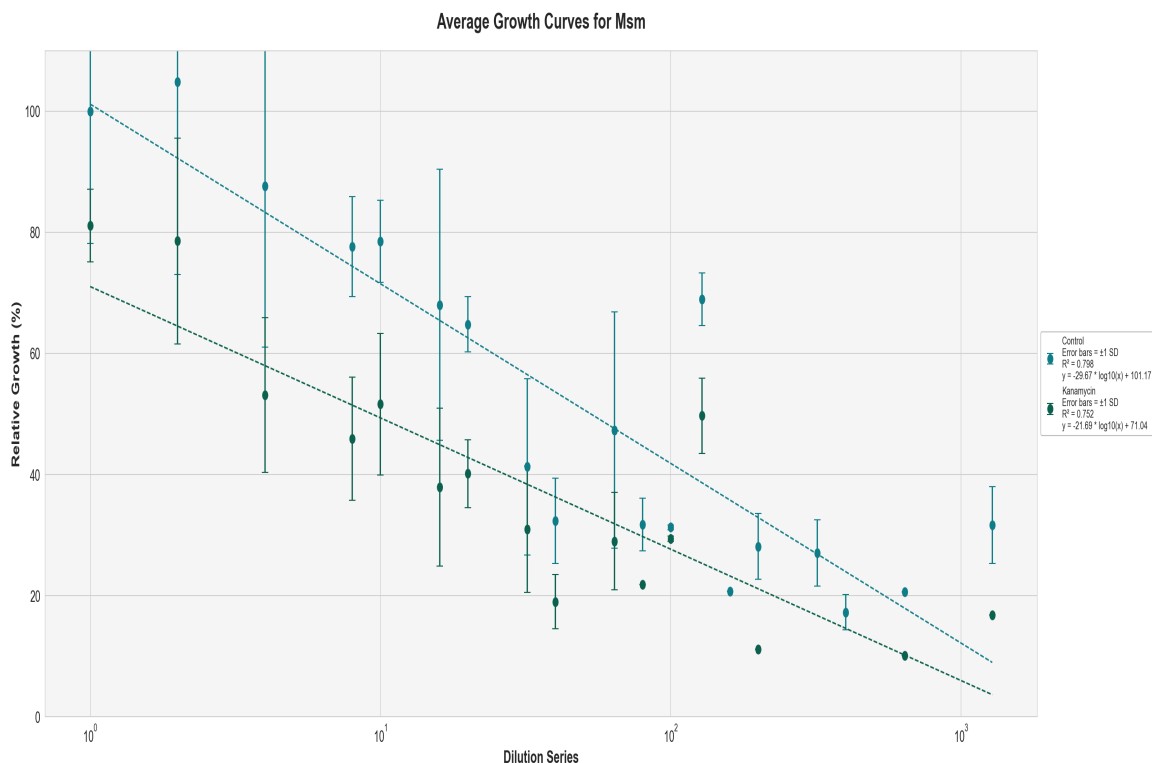
Contours Image

Individual Growth Curves for Msm



	Control (Msm-1_NoDrug_IMG_3319.JPG 0-3)	Control (Msm-1_NoDrug_IMG_3319.JPG 4-7)	Control (Msm-1_NoDrug_IMG_3319.JPG 8-11)	Kanamycin (Msm-2_Kanamycin_IMG_3323.JPG 0-3)
Formula	y = -31.77 * log10(x) + 104.91	y = -25.67 * log10(x) + 94.88	y = -27.24 * log10(x) + 99.74	y = -30.83 * log10(x) + 85.94
Slope	-31.77	-25.67	-27.24	-30.83
Intercept	104.91	94.88	99.74	85.94
R-squared	0.735	0.450	0.614	0.746
y-cut	104.91	94.88	99.74	85.94
x-cut	2006.90	4965.45	4591.43	612.98
x_at_y50	53.52	56.00	67.03	14.65

	Kanamycin (Msm-2_Kanamycin_IMG_3323.JPG 4-7)	Kanamycin (Msm-2_Kanamycin_IMG_3323.JPG 8-11)
Formula	y = -20.48 * log10(x) + 69.69	y = -18.95 * log10(x) + 64.41
Slope	-20.48	-18.95
Intercept	69.69	64.41
R-squared	0.572	0.435
y-cut	69.69	64.41
x-cut	2527.45	2501.92
x_at_y50	9.15	5.76



	Control	Kanamycin
Formula	$y = -29.67 \cdot \log_{10}(x) + 101.17$	$y = -21.69 \cdot \log_{10}(x) + 71.04$
Slope	-29.67	-21.69
Intercept	101.17	71.04
R-squared	0.798	0.752
y-cut	101.17	71.04
x-cut	2570.78	1887.01
x_at_y50	53.06	9.34

This report was generated by **Spotplotter** version **1.0** on **2024-11-21**

Spotplotter was created by **Holly Lewis** with supervision from **R Verrinder** and **Dr. M Mason** as BSc (Eng) final year project submitted in partial fulfilment of the requirements for the degree of Bachelor of Science in Electrical and

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Formula: The formula represents the linear regression equation that models determined using the `linregress` function from `scipy.stats` which determines the relationship between the logarithm of the dilution series and relative growth. It follows the form: $y = m \cdot \log_{10}(x) + b$

Normalization: The relative growth values were normalized to a baseline to make the results comparable across different conditions. The quantified values of each spot were divided by the average value of the first spot in the -ATP for each strain.

The slope (m): indicates the rate of change in relative growth as the dilution series increases (on a logarithmic scale). A steep slope indicated that the growth has a faster knockdown as the dilution changes. A shallow slope indicates that the growth is more stable across dilutions.

The intercept (b) and y-cut is the relative growth when the solution is not diluted.

The R-squared value measures how well the linear regression line fits the data, where 1 represents a perfect fit and 0 represents no relationship. Higher R^2 values indicates that the growth follows a linear relationship.

The X-cut refers to the point where the regression line crosses the x-axis, indicating the dilution value at which the relative growth would theoretically be zero (no growth).

X at Y = 50: This value represents the dilution series value when the relative growth is 50% i.e., the knockdown is 50% in comparison to the -ATP series.